

Suraksha — Complete Technology Stack

Offline-first AR industrial-safety training platform: Android worker app + web dashboard with QR-chained certificates. Version 0.1.0. No hosted backend by design.

1. Project Snapshot

Item	Detail
Product	Suraksha — worker safety training (register → train → assess → certificate → verify → export)
Version	0.1.0 (web + Android)
Platforms	Android 10+ (minSdk 29, targetSdk 35) · any modern desktop/mobile browser
Architecture	Two offline clients + JSON handoff. No server, no database host, no auth provider
Languages (UI)	English · Hindi (हिन्दी) · Santali in Ol Chiki script (ᱥᱟᱱᱛᱟᱲ)
Repository layout	android/ (Kotlin app) · dashboard/ (React web app) · docs/ · preview/ · training videos folders

2. System Architecture

Two fully-offline clients share one content + certificate contract. Data moves between them only as signed JSON export files carried on-device (Downloads folder / file import).

- Android app (Kotlin): registration, AR training (camera + 3D), assessment engine, ECDSA certificate issuance, QR display/scan, JSON site-data export.

Technology	Version	Role
React + ReactDOM	18.3.1	UI components, hooks, StrictMode
TypeScript	5.5.4 (strict, noUnusedLocals)	Type-safe code; build gate via tsc -b
Vite	5.4.2 (+ @vitejs/plugin-react 4.3.1)	Dev server (port 5173) + production bundler
Tailwind CSS	3.4.10 (+ PostCSS 8.4.41,	Utility styling; custom

		parsing
Coil (Compose)	2.7.0	Image loading
JUnit 4 / Espresso	4.13.2 / 3.6.1	Unit + instrumented tests; APK assembled by GitHub Actions CI

5. Data, Storage & Offline Model

- No backend: Supabase/hosted services were removed by decision. Zero network calls at runtime.
- Android persistence: Room (SQLite) with DAO queries incl. getByCertNo, recentEvents (AR evidence), 5-minute certificate re-issue dedupe.
- Web persistence: versioned localStorage sandbox; multi-worker registry with per-worker profiles; migration path from legacy single-

Asset	Format / location	Notes
Training modules (5)	JSON packs + assessments (android assets + dashboard public/modules)	fire_explosion, gas_confined, machinery, ppe, emergency; pass mark 80
Languages	en/hi/sat fields in every pack	Santali converted to real OI Chiki; self-hosted Noto Sans OI Chiki woff2 (400/600/700, ~15 KB each)
3D models (Android AR)	6 × .glb in assets/models	extinguisher, safety sign, gas

		cylinder, helmet, machinery guard, first-aid kit
Simulation videos	MP4 H.264 1280×720 (~10 s) in public/videos	Gemini clips for fire/gas/gear/emergency, sectioned per step; ffmpeg-made branded cards for PPE
Illustrations/icons	Inline SVG + public/img/*.svg	3D-style shield logo, module thumbnails, hero banner, favicons
Fonts (Android)	Noto Sans Ol Chiki TTF in res/font	Ol Chiki renders on any Android 10+ device

8. Design System

Token	Value	Usage
Deep Navy #0B2545	brand DEFAULT	Primary brand, headers, primary surfaces
Steel Blue #1B4965	brand-600	Secondary headers/nav
Terracotta #C97B5A	hazard DEFAULT	All CTA buttons
Muted Gold #D9A566	caution DEFAULT	Secondary highlights, badges
Sage #7A9B76	success DEFAULT	Passed / VALID states
Clay Red #C17B72	alert DEFAULT	Failed / INVALID states
Charcoal #2D3436 / Light Gray #F5F6F8 / White	ink / paper / cards	Body text, page background, cards

- Effects: card lift + deep shadows on hover, image zoom, button press physics, shine sweeps, pulsing checkpoints, animated progress bars.
- Per-module accent colors (fire orange, gas sky, machinery steel, PPE amber, emergency red) on cards, banners and lesson dividers.

9. DevOps, QA & Tooling

- CI: .github/workflows/android-ci.yml assembles the debug/release APK (no local JDK/Android SDK on dev machine).
- Gates: tsc

- Dev servers: Vite on :5173 (binds localhost + LAN); Chrome used for manual verification of every flow.

10. Key Source Paths

Area	Path
Web entry / routes / shell	dashboard/src/main.tsx, App.tsx, components/Layout.tsx
Worker app	dashboard/src/pages/train/ (TrainHome, TrainModule, TrainAssess, TrainCert)
3D + gestures + voice	dashboard/src/components/ (Stage3D, StepStage, CameraSimulation, Info3D, simGestures)
Trust + i18n + stores	dashboard/src/lib/ (chain, verify, content, i18n, voice, workerStore, store)
Android app	android/app/src/main/java/com/suraksha/trainer/ (ui, cert, data, assessment, ar)
Android resources	android/app/src/main/{assets/{modules,models}, res/{values*, drawable, mipmap-*, font}}
CI	.github/workflows/android-ci.yml

Generated from the live codebase. Native-speaker review of Santali strings is still required before production use with workers.